

Physics 7501 Quiz 2
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(1.2.1) Pauli matrices

What are the eigenvalues of the 2×2 matrix $e^{i\theta\hat{Z}}$ where $\hat{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ is the 3rd Pauli matrix?

For a diagonal matrix M , we exponentiate every element along the main diagonal to get

$$e^M = \begin{pmatrix} e^{M_{11}} & & 0 \\ & \ddots & \\ 0 & & e^{M_{nn}} \end{pmatrix}$$

thus

$$e^{i\theta\hat{Z}} = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}$$

Since this is a diagonal matrix, its eigenvalues are just the diagonal entries $e^{\pm i\theta}$

(1.2.2) Linear algebra

Find the simultaneous (i.e. common) eigenkets of Pauli matrices $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.
If you cannot find any, explain.

σ_x and σ_z have no simultaneous eigenkets. We know from homework that non-commuting Hermitian operators cannot share a common *basis* of eigenkets. In this case however, we can make the stronger statement that they share *none*. Suppose they shared a non-zero eigenket $|\psi\rangle$

$$\begin{aligned} [X, Z]|\psi\rangle &= (XZ - ZX)|\psi\rangle \\ &= xz|\psi\rangle - zx|\psi\rangle \\ &= 0 \end{aligned}$$

$\implies$

$$[X, Z]|\psi\rangle = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -2a \\ 2b \end{pmatrix} \implies a = b = 0$$

which contradicts our original presumption that $|\psi\rangle$ was non-zero.

(1.2.3) Measurements

Consider the following normalized state of a single qubit:

$$|\psi\rangle = \begin{pmatrix} \cos \theta e^{i\phi} \\ \sin \theta \end{pmatrix}. \quad (1.3)$$

For a measurement of Pauli operator $\hat{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, what is the probability $\mathbf{P}(X = +1)$ of measurement outcome $X = 1$? Express it as a function of θ and ϕ .

In the standard $\hat{Z}$ -basis, the $X = +1$ state is given by

$$|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

The Born rule states that $\mathbf{P}(X = +1) = |\langle + | \psi \rangle|^2$, thus

$$\langle + | \psi \rangle = \frac{1}{\sqrt{2}} (\cos \theta e^{i\phi} + \sin \theta)$$

$\Rightarrow$

$$\begin{aligned} \mathbf{P}(X = +1) &= |\langle + | \psi \rangle|^2 = \frac{1}{2} (\cos^2 \phi + (e^{-i\phi} + e^{i\phi}) \sin \theta \cos \theta + \sin^2 \theta) \\ &= \frac{1}{2} (1 + 2 \sin \theta \cos \theta \cos \phi) \\ &= \frac{1}{2} (1 + \sin(2\theta) \cos \phi) \end{aligned}$$